

Motivation. Most continual learning (CL) methods assume supervised labels and focus on preserving task-specific decision boundaries. However, real-world data streams are often unlabeled and non-stationary. This paper studies *unsupervised continual learning (UCL)*, where the objective is to learn feature representations sequentially without catastrophic forgetting.

Core Hypothesis. Unsupervised representations are inherently more robust to forgetting because they learn general-purpose features rather than task-specific discriminative boundaries.

Methodology. The authors adapt modern self-supervised learning methods—**SimSiam** and **Barlow Twins**—to the continual learning setting. Representations are trained sequentially across tasks and evaluated using a frozen encoder with a KNN classifier.

They extend several supervised CL strategies to UCL:

- Regularization (Synaptic Intelligence)
- Architectural expansion (Progressive Neural Networks)
- Rehearsal (Dark Experience Replay)

They additionally propose **Lifelong Unsupervised Mixup (LUMP)**, which interpolates current-task examples with replay-buffer samples:

$$\tilde{x}_{i,\tau} = \lambda x_{i,\tau} + (1 - \lambda)x_{j,M}$$

This encourages linear behavior across tasks and mitigates representational drift.

Key Findings.

- UCL significantly reduces catastrophic forgetting compared to supervised CL.
- Simple finetuning in UCL often outperforms sophisticated supervised CL methods.
- UCL generalizes better to out-of-distribution (OOD) tasks.
- UCL is more robust in few-shot settings.
- Loss landscapes of UCL are flatter and smoother than supervised CL.

Representational Analysis. Using CKA similarity and ℓ_2 parameter distance:

- Two independent UCL models are more similar to each other than two supervised models.
- Lower layers remain similar across methods; higher layers diverge.
- UCL representations produce coherent, perceptually meaningful feature maps even for earlier tasks.

Conclusion. Unsupervised continual learning yields more stable, transferable representations than supervised CL. Learning invariant, general-purpose features implicitly regularizes the learning dynamics and reduces forgetting.